Data Science Collective

Stylometric Watermarks vs. LLM Watermarks: Can We Really Trace AI Authorship?

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Introduction: The Push to Identify AI-Generated Text

In an era where AI-written prose can mimic human style with alarming fidelity, distinguishing human authorship from machine generation has become a pressing challenge. Content platforms are starting to respond.

For example, Medium announced in 2024 that fully AI-generated stories would be restricted: even if disclosed, they'd only get basic algorithmic distribution (no human curation boosts), and undisclosed AI text would be shown *only* to the author's followers.

Underlying such policies is a growing concern: **Can we reliably trace AI authorship in text?**

Two promising approaches have emerged:

Stylometric analysis and **watermarking**, each attempting to tag or reveal AI-written content.

This article explores what they are, how they differ, and whether they truly let us say, "Aha, an AI wrote this!" with confidence.

Stylometry: Every Writer Has a Fingerprint

Stylometry is the science of analyzing writing style to identify authorship.

The premise is simple: each person writes with unique quirksfrom favorite words and punctuation habits to sentence structure and syntax.

These subtle patterns form a sort of literary fingerprint.

Stylometry has a storied history in human authorship attribution.

It's been used in court cases (it helped **convict the Unabomber** by matching writing style) and in plagiarism detection.

If stylometry can unmask a pseudonymous human author, could it also detect when an *AI* is the author?

The answer is complicated.

Traditional stylometry methods look at vocabulary richness, average sentence length, use of function words, etc., to profile an author. Early studies did attempt to apply this to AI-generated text.

In some cases, rudimentary models like GPT-2 did exhibit telltale stylistic oddities that a well-trained classifier could pick up.

However, as large language models (LLMs) have evolved, their outputs have become more human-like and varied.

In fact, researchers found that classic authorship attribution techniques struggled to reliably distinguish even GPT-2's writing from human text.

With GPT-3 and beyond, it's even harder *for humans to tell AI text apart*, and likewise challenging for stylometric algorithms.

Stylometric watermarks (style-based clues):

One recent idea is to *flip stylometry on its head*. Instead of merely analyzing style post-hoc, some researchers propose embedding a **stylometric watermark** *during* text generation. In 2023, AI experts suggested that LLMs could intentionally adopt a particular stylistic pattern as a hidden signal.

For example, an AI model might consistently favor an unusual sentence structure or always choose a certain synonym over alternatives that are differences so subtle that readers won't notice, but statistically rare in human writing.

Because these patterns are deliberate, they act like an invisible signature woven into the prose.

A detection algorithm (aware of the intended pattern) could later flag content that carries this stylometric fingerprint. In essence, it's watermarking via writing style.

One experiment in this vein introduced **acrostics and "sensorimotor" word choices** as watermarks in AI text.

The AI would, say, start each sentence such that the first letters spell out a message (an acrostic), or preferentially use words from a certain sensory category ("smell" words instead of "sight" words, for instance). These choices are controlled by a secret *key* that changes every sentence, making the pattern complex yet detectable.

The resulting Stylo metric quirks are too consistent to be accidental.

In tests, this method allowed the detection of AI authorship with high confidence for texts just a few sentences long.

Crucially, it didn't require fine-tuning a model or using an external AI to detect; instead, simple statistical checks sufficed to verify the presence of the watermark.

Limitations of stylometric approaches:

Stylometry-based detection, whether traditional or watermark-based, faces an arms race.

An AI *without* a watermark can purposely vary its style or even be instructed to mimic a human writing quirk, reducing the effectiveness of straightforward stylometric checks.

On the flip side, an AI *with* a stylometric watermark could have it scrubbed out by post-editing.

Imagine a person paraphrasing each sentence or swapping in different synonyms, this could neutralize the patterns the watermark tried to embed.

In fact, tools already exist to perform "stylometric obfuscation", i.e. automatically rephrase text to erase stylistic fingerprints.

Stylometry also runs the risk of false positives: a human author might coincidentally exhibit an unusual style that looks "AI-like" or even accidentally trigger an AI's watermark pattern.

Such cases are rare, but they underscore why stylometric signals alone can't be absolute proof.

LLM Watermarks: Hiding a Signal in the Tokens

Another leading strategy to trace AI-generated text is through **LLM watermarks** that operate at the token (word) level.

These are typically *invisible, statistical signals* inserted into the text by the AI model as it generates output. Unlike a visible watermark on an image (like a translucent logo), LLM watermarks are designed to blend into natural language, only detectable through algorithmic analysis.

The core idea is to bias the AI's word choice in a subtle way that doesn't degrade the text quality but does leave a measurable imprint in the distribution of words.

How does this work in practice? One prominent approach is often described as a "green list vs. red list" method (see *Figure 1* below).

Before the AI begins generating, it uses a secret random seed to label certain tokens (words or subwords) as *preferred* (green) and others as *avoided* (red).

As the AI writes each sentence, whenever it has some freedom in word choice, it leans toward using a green-listed word if it makes sense, and steers away from red-listed words.

The effect is that over a long passage, an unusually high frequency of specific "green" tokens will appear.

In turn, that is not enough to seem repetitive, but enough that statistically it's very unlikely to be a coincidence.

A verifier with knowledge of the secret seed can count the occurrence of those tokens and compute a score (like a z-score) to judge if the pattern of greens is too strong to have occurred naturally.

This method, introduced by researchers in late 2022 and 2023, is essentially a **cryptographic watermark** for text: only someone with the secret key knows which words to check, and the watermark is "zero-bit" (its presence just indicates AI origin, though more complex versions can encode multi-bit messages).

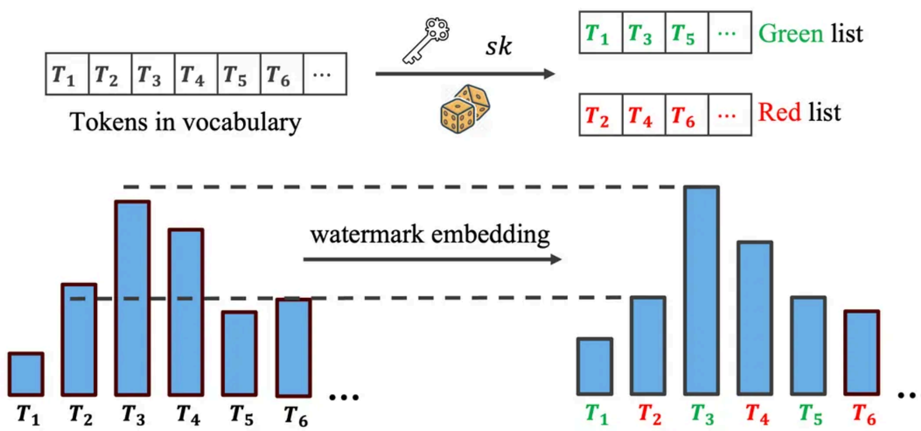

Figure 1. An illustration of an LLM watermarking scheme (green-list method). The model's vocabulary is partitioned into "green" tokens (preferred) and "red" tokens (avoided) using a secret key. During generation, the AI subtly biases its next-word selection towards green tokens. The resulting text looks normal to a reader, but statistically, the distribution of tokens (bottom right) shows greens appearing more often than random chance. A detector with the secret key can measure this bias to confirm AI authorship.

This kind of LLM watermark can be quite robust in theory.

Research shows that with a long enough passage (hundreds of tokens), detectors can catch the watermark with over **97% true-positive accuracy** while maintaining high text quality (fluency).

Even a short snippet (e.g. a paragraph of 100 tokens) can often be flagged with statistical confidence if it's watermarked.

Companies have taken notice: by late 2024, Google's DeepMind announced a production-ready text watermark (part of their **SynthID** system) that was even published in *Nature*.

Amazon's Bedrock service likewise introduced a detectable watermark for images and hinted at a similar one for text.

Clearly, the technology is maturing.

The cat-and-mouse game: However, just as with stylometry, watermarks face adversaries.

The simplest attack is **paraphrasing** where you take a watermarked text and have either a human or another AI rewrite it in different words.

This tends to scramble the specific token frequencies and can destroy the hidden signal. In fact, OpenAI's own researchers (like Scott Aaronson) acknowledged that a determined user could defeat their prototype watermark by rephrasing the output or swapping words with synonyms.

Empirically, soon after academic proposals for token-based watermarks came out, community hackers built tools to do exactly that: automatically replace words, shuffle phrases, or use back-translation (translating to another language and back) to shake off watermarks.

Because the watermark is designed to be invisible and not ruin the text, it often doesn't take much editing to remove it without changing the meaning.

Watermark researchers are not sitting idle, of course. They've introduced measures like **error-correcting codes and robust encoding** to make watermarks survive some level of paraphrasing or noise.

For instance, a watermark might be redundantly embedded across many parts of the text, so that even if a few sections get reworded, enough of the signal remains to detect. Some designs operate at the level of syntax or sentence structure rather than exact word choice, making them harder to erase without significant rewritten.

Nonetheless, it's a never-ending chess match: as one paper's title well put it, there's "No Free Lunch in LLM Watermarking"; every design trade-off (like making a watermark robust) can open a new vulnerability (like an exploit to *spoof* the watermark).

We're witnessing an evolving arms race between watermark developers and attackers attempting to evade or even falsely trigger them.

Tracing AI Authorship: Are We There Yet?

Despite the challenges, these approaches mark significant steps toward traceable AI content.

Stylometric analysis leverages the inherent "fingerprint" in writing, and watermarking actively plants a detectable signature.

Together, they approach the problem from both ends: one *passive* (observe the style) and one *active* (embed a code).

So, can we really trace AI authorship now? The answer is **partly yes**, but with important caveats:

- **Reliability vs. Evasion:** For honest use-cases, where an AI system voluntarily watermarks its output, detection can be highly reliable under ideal conditions. A very long article generated by a compliant AI can indeed be flagged with near-certainty by automated checks. However, if someone is *intentionally trying to hide* AI authorship, they have a toolkit of tricks to defeat detection, ranging from paraphrasers that break watermarks to style-transfer techniques that confound stylometry. Tracing authorship then becomes much harder, often requiring the investigation of multiple clues (or direct access to the AI model's logs, if available).

- **False Positives/Negatives:** No current method is foolproof. A short AI-generated snippet (e.g. a one-sentence answer) may evade a watermark detector simply because there wasn't enough text to imprint a clear signal. Conversely, purely human-written text might occasionally trigger a naive AI-detector if it coincidentally has odd statistics. (Notably, OpenAI's own AI-written text classifier was withdrawn in 2023 for being too inaccurate, underscoring how hard general detection is.) Watermarks grounded in cryptography have clear statistical thresholds to minimize false alarms, but stylometric clues can be fuzzier. Curation teams like Medium's explicitly avoid distributing stories that read as low-quality or spammy but this is a proxy and not a true authorship test.
- **Ethical and Practical Considerations:** Implementing widespread AI traceability raises its own questions. There's a balance between **accountability** (knowing when AI was used, to prevent misuse or fraud) and **privacy/anonymity**. If every AI model watermarks its output, should users have the right to remove or obscure it for legitimate reasons? Also, detection needs to be handled carefully, as such, an accusation of "you didn't write this" can be sensitive. The goal of these technologies is to provide *evidence* and assurance, not to cast doubt randomly. As one AI policy observer noted, watermarking is meant as a safety net for transparency and trust, not a hard prohibition.

Closing Thoughts

Tracing AI authorship is no longer science fiction; it's an active field of research and product development.

Stylometric watermarks and LLM token watermarks each offer pieces of the puzzle.

We can often tell if *a text was AI-generated* when using these methods, especially if the AI played along and the text is substantial.

What's harder is attributing *which* AI or ensuring the trace survives adversarial editing.

In 2025, the state of the art is that AI content can be tagged and detected under favorable conditions, but determined actors can still slip through the net.

In other words, we have new **tools for transparency** but not an absolute guarantee.

As AI continues to write more of the content we consume, expect this cat-and-mouse game to continue with tighter watermarks, smarter detectors, and yes, cleverer counter-measures.

We're getting better at spotting AI-generated content, but the investigation behind the scenes is far from over.

If you found this exploration useful, consider sharing it or joining the conversation: how do you think we should balance transparency, privacy, and accountability in the age of AI authorship?

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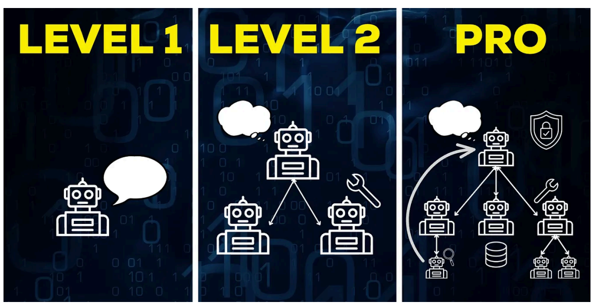

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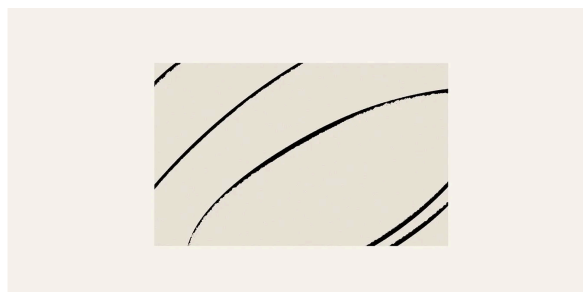

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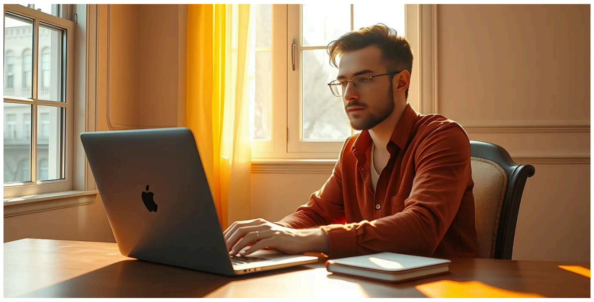

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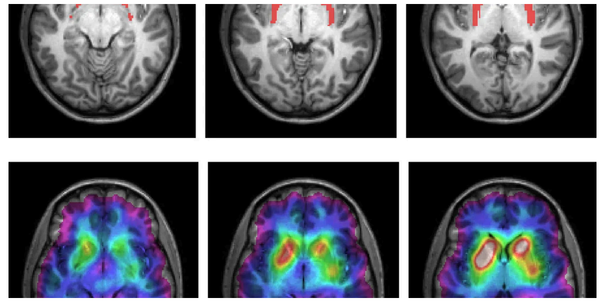

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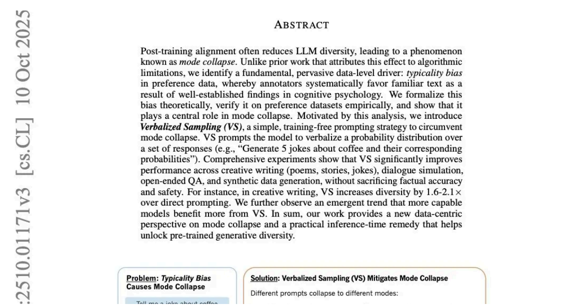
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Abstract

Post-training alignment often reduces LLM diversity, leading to a phenomenon known as *mode collapse*. Unlike prior work that attributes this effect to algorithmic limitations, we identify a fundamental, pervasive data-level driver: *typicality bias* in preference data, whereby annotators systematically favor familiar text as a result of well-established findings in cognitive psychology. We formalize this bias theoretically, verify it on preference datasets empirically, and show that it plays a central role in mode collapse. Motivated by this analysis, we introduce ***Verbalized Sampling (VS)***, a simple, training-free prompting strategy to circumvent mode collapse. VS prompts the model to verbalize a probability distribution over a set of responses (e.g., "Generate 5 jokes about coffee and their corresponding probabilities"). Comprehensive experiments show that VS significantly improves performance across creative writing (poems, stories, jokes), dialogue simulation, open-ended QA, and synthetic data generation, without sacrificing factual accuracy and safety. For instance, in creative writing, VS increases diversity by 1.6-2.1× over direct prompting. We further observe an emergent trend that more capable models benefit more from VS. In sum, our work provides a new data-centric perspective on mode collapse and a practical inference-time remedy that helps unlock pre-trained generative diversity.

Problem: *Typicality Bias* Causes Mode Collapse

Solution: Verbalized Sampling (VS) Mitigates Mode Collapse

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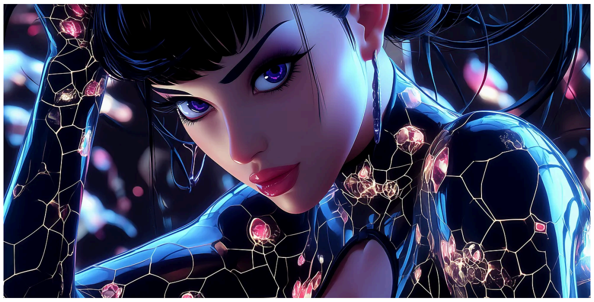

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